MOHAMMED ABDULLAH

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SUMMARY

Mechanical engineer who has worked in quality at Bosch and in hydraulic cylinder design at Minar Industries. Certified SOLIDWORKS Professional (CSWP) seeking a co-op in mechanical design, manufacturing, or quality engineering.

EDUCATION

Northeastern University, Boston, MA — Expected May 2028
Master of Science in Mechatronics, Robotics, and Automation, Concentration in Mechanical Engineering
Relevant Coursework: Robotics Sensing and Navigation, Mobile Robotics, Robot Design, CAD and Manufacturing

Visvesvaraya Technological University, Karnataka, India — Jun 2023
Bachelor of Engineering in Mechanical Engineering

TECHNICAL SKILLS

CAD & Software: SOLIDWORKS, CATIA, Siemens NX, AutoCAD, Ansys (FEA), Revit MEP, Microsoft Office
Quality & Manufacturing: Micrometers, precision gauges, gauge calibration, GD&T, FMEA, root cause analysis, CAPA, ISO 9001, lean manufacturing (5S), Six Sigma, CNC machining (lathe and milling)
Programming & Robotics: Python, C programming, Claude Code, ROS 2, Linux (Ubuntu), Git, IMU and GPS data

PROFESSIONAL EXPERIENCE

Design Engineer, Minar Industries, Karnataka, India — Jun 2025 - Oct 2025
- Designed hydraulic cylinders and pistons for heavy machinery and took each one from concept through design review
- Prepared the manufacturing drawings, assembly drawings, and bills of materials (BOMs) used to build each part
- Verified part dimensions, tolerances, and clearances against engineering and performance standards before production
- Partnered with production, quality, and supplier teams to make manufacturing more efficient and parts more reliable

Graduate Engineer, Bosch, Karnataka, India — Jul 2023 - May 2025
- Worked in a precision diesel component service and manufacturing environment, managing quality control, production, and assembly operations in compliance with ISO 9001 standards
- Applied FMEA and root cause analysis to trace quality problems, then put corrective and preventive actions (CAPA) in place to reduce defects
- Inspected and tested diesel parts with calibrated gauges, checked gauge accuracy with MSA, and logged results in SAP
- Contributed to a 15% improvement in test efficiency by developing standardized testing procedures and reporting quality metrics; recognized with the Impactful Engineering Award
- Supported lean workshop improvements, including 5S implementation and safer work practices

PROJECTS

Terrain Classification from Wheel-Mounted IMU Vibrations — Jan 2026 - Apr 2026
Northeastern University, Boston, MA
- Mounted a VectorNav VN-100 IMU on the wheel shaft of a Mecabot Plus robot to sense floor vibrations
- Recorded 40 test runs on tile, hardwood, carpet, and grass with ROS 2 on Linux, producing 3,474 labeled data windows
- Classified the four surfaces in Python with 87.3% test accuracy and gave each its own Xbox controller vibration in Unity

Mini Robotic Cart Design — Jan 2023 - May 2023
Visvesvaraya Technological University, Karnataka, India
- Modeled a mini robotic cart chassis in SOLIDWORKS, including the frame, wheel mounts, and motor housings
- Arranged the layout for balance and easy assembly, with all parts ready to fabricate

Renewable Energy-Powered Water Purification System — Aug 2022 - Dec 2022
Visvesvaraya Technological University, Karnataka, India
- Developed a brine water purification system powered by solar and wind energy
- Added sensors and automatic valve, pump, and motor control so the system needed less manual work

CERTIFICATIONS, TRAINING & ACTIVITIES

Building Practical Python Programming Skills for Technology Careers, LinkedIn Learning — In progress
Certified SOLIDWORKS Professional (CSWP) and Associate (CSWA), Dassault Systèmes — Apr 2026
CNC Machining (Lathe and Milling) Training, Karnataka German Technical Training Institute, India — Apr 2024
Member, American Society of Mechanical Engineers (ASME), NUHOC Member Northeastern University — Jan 2026 - Present